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%%Module: SME3701
%% Date: 28-04-2026
%%Git hub link:https://github.com/ndivhenijamex-web/18813143

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%%Undamped Free vibration of an SDOF mass system

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clc; clear; close all;
%data given
m =5;
k =2000;
x0= 0.02;
v0= 0;

%%Q1_1 Natural angular frequency
wn =sqrt (k/m);
fprintf('Natural Angular frequency=%.2frad/s\n',wn);

```

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Natural Angular frequency=20.00rad/s

```

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%%Q1_2 Displacement Equation
wn= sqrt (k/m);
t= 0:0.01:5;
%Displacement Equation
x = x0 * cos(wn * t);
disp ('Displacement values x (t):');

```

```

Displacement values x (t):

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disp (x);

```

```

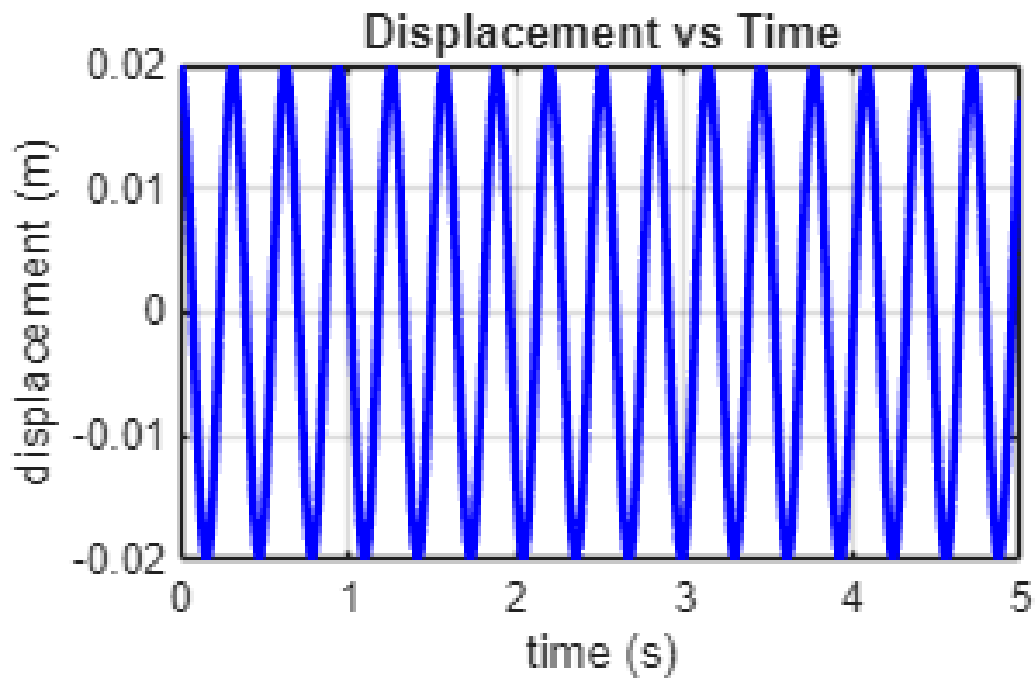
0.0200    0.0196    0.0184    0.0165    0.0139    0.0108    0.0072    0.0034   -0.0006   -0.0045   -0.0083   -0

```

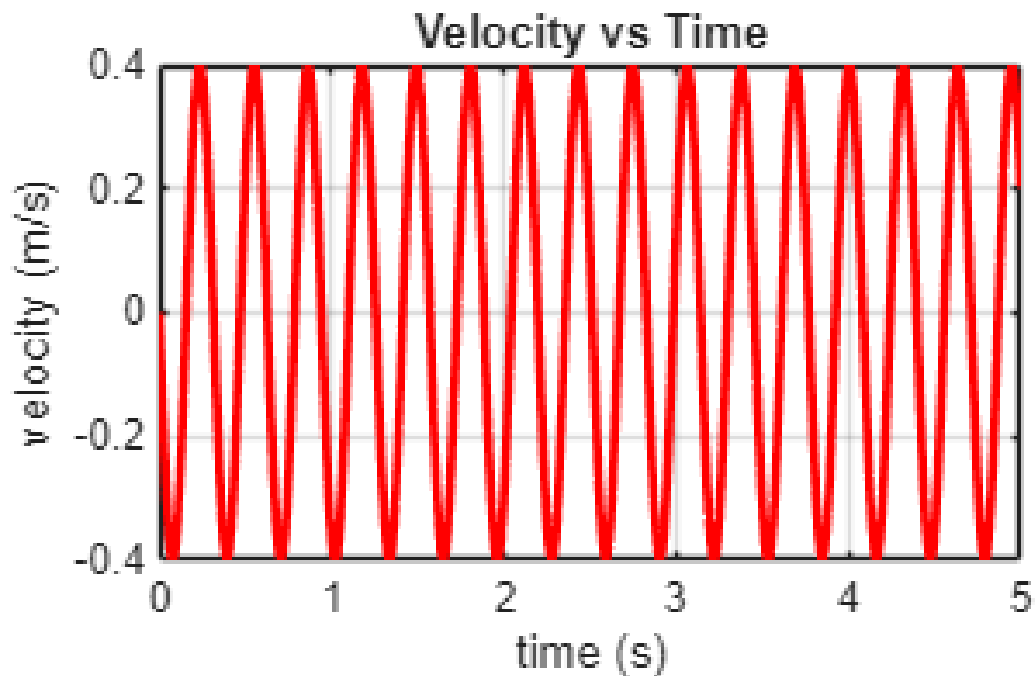
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%%Q1_3 Displacement plot
wn =sqrt (k/m);
t= 0:0.01:5;
x = x0* cos (wn*t);
plot (t,x,'b','linewidth',2);
grid on;
xlabel ('time (s)');
ylabel('displacement (m)');
title ('Displacement vs Time'),

```



```
%Q1_4 Velocityplot  
wn =sqrt (k/m);  
t= 0:0.01:5;  
% Velocity Equation  
v = -x0 * wn * sin(wn * t);  
plot (t,v,'r', 'linewidth',2);  
grid on;  
xlabel('time (s)');  
ylabel('velocity (m/s)');  
title('Velocity vs Time');
```



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%%Q1_5 Comments
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disp ('nature of motion: simple hamonic');
```

```
nature of motion: simple hamonic
```

```
disp ('zero damping meaning no energy loss');
```

```
zero damping meaning no energy loss
```

```
disp ('velocity is 90 degrees out of phase with displacement');
```

```
velocity is 90 degrees out of phase with displacement
```

```
disp ('Velocity is maximum at zero displacement');
```

```
Velocity is maximum at zero displacement
```